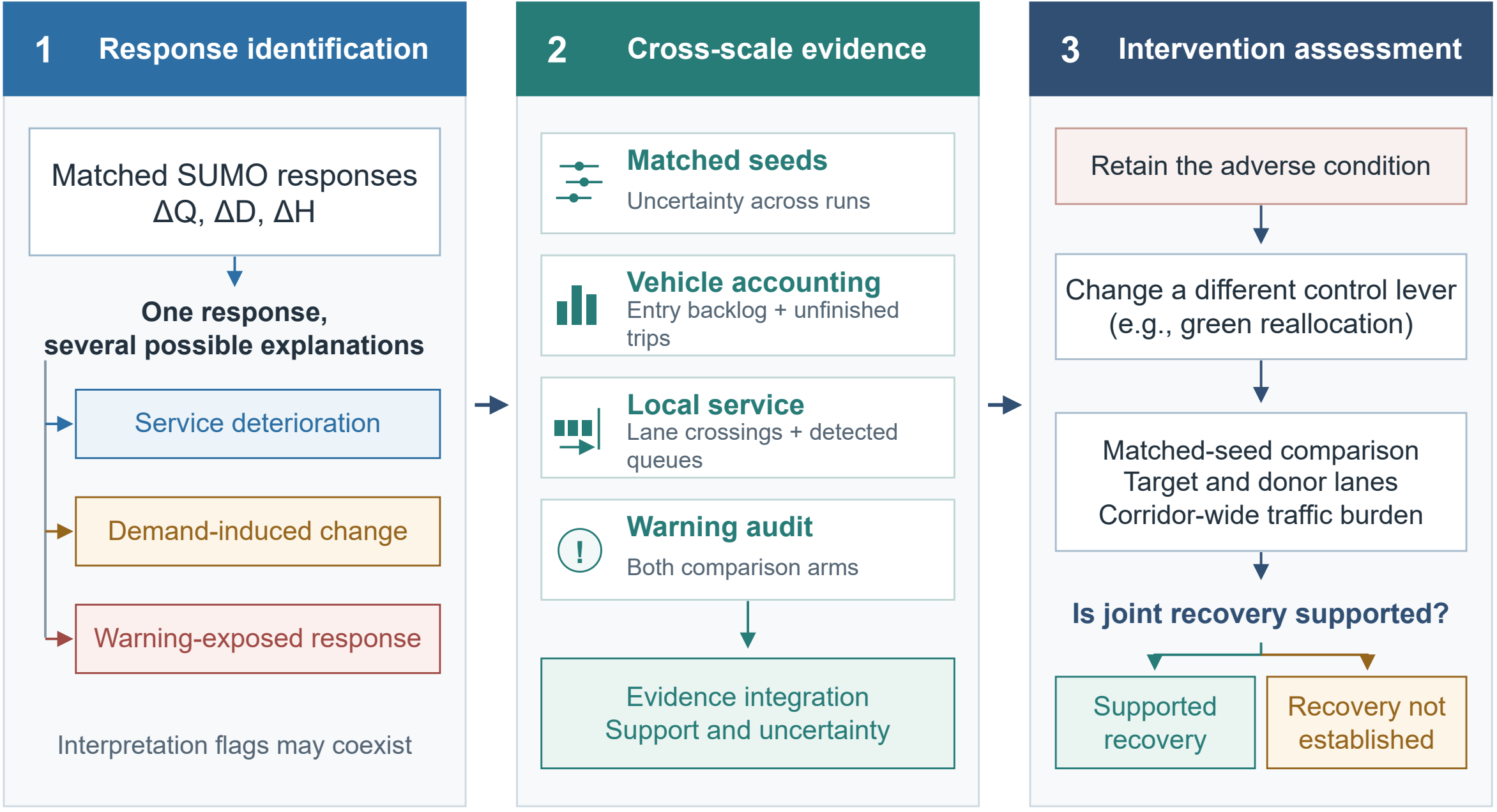
1 Response identification
Matched SUMO responses
ΔQ, ΔD, ΔH
One response,
several possible explanations
Service deterioration
Demand-induced change
Warning-exposed response
Interpretation flags may coexist
2 Cross-scale evidence
Matched seeds
Uncertainty across runs
Vehicle accounting
Entry backlog + unfinished trips
Local service
Lane crossings + detected queues
Warning audit
Both comparison arms
Evidence integration
Support and uncertainty
3 Intervention assessment
Retain the adverse condition
Change a different control lever
(e.g., green reallocation)
Matched-seed comparison
Target and donor lanes
Corridor-wide traffic burden
Is joint recovery supported?
Supported recovery
Recovery not established